From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2013

FULL TEST – II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 252**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two section: **Section-A & Section-C**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) Section-A (01 – 08) contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 13) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

Section-A (14 – 18) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) Section-C (01 – 10) contains 10 Numerical based questions with answers as numerical value and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

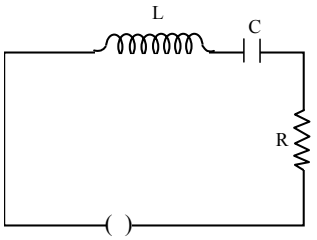
SECTION – A
Single Correct Choice Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A circuit has a self inductance of 1 henry and carries a current of 2A. To prevent sparking when the circuit is broken, a capacitor which can withstand 400 volts is used. The least capacitance of the capacitor connected across the switch must be equal to
   (A) 12.5 μF (B) 25 μF
   (C) 50 μ F (D) 100 μF

2. A spherical conductor has a radius of R and charge Q. A spherical shell of thickness R and uniform charge Q, is kept so as to be concentric to the conductor and touching the conductor. The electric field at a distance of 1.5 R from the center of the sphere is
   (A) $\frac{83Q}{504\pi\varepsilon_0 R^2}$ (B) $\frac{75Q}{504\pi\varepsilon_0 R^2}$
   (C) $\frac{83Q}{252\pi\varepsilon_0 R^2}$ (D) $\frac{53Q}{252\pi\varepsilon_0 R^2}$

3. Figure shows a series LCR circuit connected to a variable frequency 200V source. L = 5H, C = 80 μF and R = 40 Ω. What is the source frequency which drives the circuit at resonance?
   (A) 25 Hz (B) 25/π Hz
   (C) 50 Hz (D) 50/π Hz

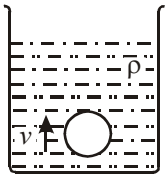


Rough work

4. A particle of mass m moves towards a smooth vertical wall with a speed u (relative to the ground) and collides elastically with the wall; the wall moving towards the particle with a speed 2u (also relative to the ground). Assuming that the wall is extremely massive, the impulse delivered to the particle equals

(A) 5 mu (B) 6 mu
(C) 4 mu (D) none of these.

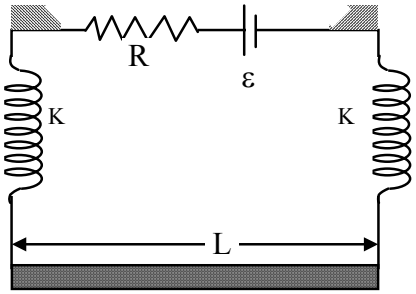
5. An air bubble of radius r rises steadily through a liquid of density ρ at the rate of v. Neglecting density of air, the coefficient of viscosity of liquid is



(A) $\frac{2}{9}\frac{r^2\rho g}{v}$ (B) $\frac{1}{3}\frac{r^2\rho g}{v}$

(C) $\frac{1}{9}\frac{r^2\rho g}{v}$ (D) None of these

6. A perfectly conducting straight rigid rod of mass m and length L is suspended symmetrically from two identical perfectly conducting springs as shown in the figure. The springs stretch a distance of x_0 due to the weight of the wire. The circuit has a total resistance of RΩ. When the magnetic field perpendicular to the plane of the paper is switched on, springs are observed to extend further by the same distance. The magnetic field strength is



(A) $\frac{mgR}{\varepsilon L}$; directed outward from the plane of the paper

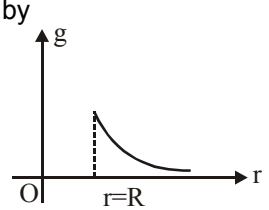
(B) $\frac{mgR}{2\varepsilon x_0}$; directed outward from the plane of the paper

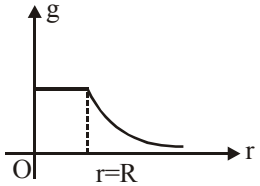
(C) $\frac{mgR}{\varepsilon L}$; directed into the plane of the paper

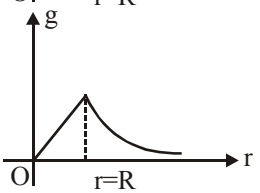
(D) $\frac{mgR}{\varepsilon x_0}$; directed into the plane of the paper

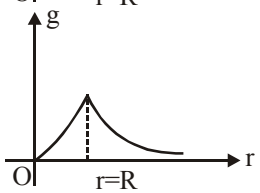
Rough work

7. The dimension of $\varepsilon_0 E^2$ (ε_o : permittivity of free space, *E* : electric field, *J* : dimension of energy) (Choose the most appropriate option)
(A) $J L^{-3}$ (B) $J L^{-1} T^{-2}$
(C) $ML^{-1}T^{-2}$ (D) (A) & (C)

8. The variation of gravitational intensity g for a uniform solid sphere of mass M and radius R is given by
(A) 

(B) 

(C) 

(D) 

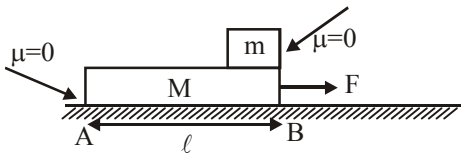
Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

9. A particle moves along positive branch of the curve $y = \frac{x}{2}$ where $x = \frac{t^3}{3}$, x and y are measured in metres and t in seconds, then
(A) the velocity of particle at t = 1 s is $\hat{i} + \frac{1}{2}\hat{j}$
(B) the velocity of particle at t = 1 s is $\frac{1}{2}\hat{i} + \hat{j}$
(C) the acceleration of particle at t = 1 s is $2\hat{i} + \hat{j}$
(D) the acceleration of particle at t = 2 s is $\hat{i} + 2\hat{j}$

Rough work

10. In the figure small block is kept on m then



(A) the acceleration of m w.r.t. ground is $\frac{F}{m}$

(B) the acceleration of m w.r.t. ground is zero

(C) the time taken by m to separate from M is $\sqrt{\frac{2\ell m}{F}}$

(D) the time taken by m to separate from M is $\sqrt{\frac{2\ell M}{F}}$

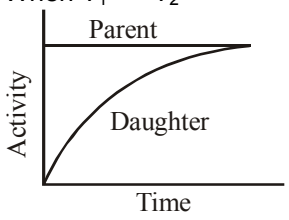
11. When a current–carrying coil is placed in a uniform magnetic field with its magnetic moment anti–parallel to the field.

(A) Torque on it is maximum (B) Torque on it is zero

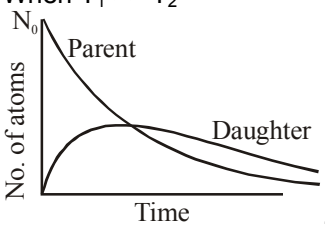
(C) Potential energy is maximum (D) Dipole is in unstable equilibrium.

12. A nucleus A decays into B with half life T_1 and B decays into C with half life T_2. Graph is drawn between number of atoms/activity versus time. Select the correct graph(s)

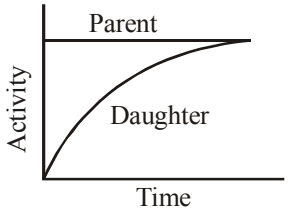
(A) When $T_1 >> T_2$



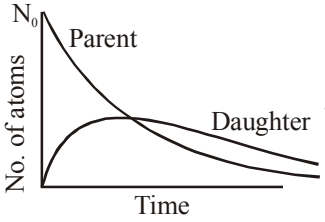
(B) When $T_1 >> T_2$



(C) When $T_1 < T_2$



(D) When $T_1 < T_2$



13. A lens of focal length 'f' is placed in between an object and screen at a distance 'D'. The lens forms two real images of object on the screen for two of its different positions, a distance 'x' apart. The two real images have magnifications m_1 and m_2 respectively ($m_1 > m_2$). Then

(A) $f = \frac{x}{m_1 - m_2}$ (B) $m_1 m_2 = 1$

(C) $f = \frac{D^2 - x^2}{4D}$ (D) $D \geq 4f$.

Rough work

PARAGRAPH TYPE

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph for Question Nos. 14 to 15

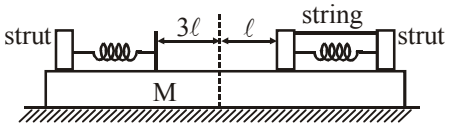
A coil having a resistance of 50.0Ω and an inductance of 0.5 H is connected to an AC source of 110 V, 50 Hz.

14. The impedance of the coil is (approximately)
(A) 165Ω (B) 55Ω
(C) 330Ω (D) none of these

15. The rms current in the circuit is
(A) 1.5 A (B) 0.67 A
(C) 1 A (D) 3 A

Rough work

Paragraph for Question Nos. 16 to 18



According to the principle of conservation of linear momentum if the external force acting on the system is zero, the linear momentum of the system will remain conserved. It means if the centre of mass of a system is initially at rest, it will remain at rest in the absence of external force, that is the displacement of centre of mass will be zero.

A plank of mass M is placed on a smooth horizontal surface. Two light identical springs each of stiffness K are rigidly connected to struts at the end of the plank as shown in the figure. When the springs are in their unextended position the distance between their free ends is 3ℓ. A block of mass m is placed on the plank and pressed against one of the springs so that it is compressed to ℓ. To keep the block at rest it is connected to the strut by means of a light string. Initially the system is at rest. Now the string is burnt.

16. The maximum displacement of plank is

(A) $\frac{5m\ell}{M}$ (B) $\frac{5m\ell}{M+m}$

(C) $\frac{3m\ell}{M+m}$ (D) $\frac{4m\ell}{M+m}$

17. Maximum velocity of plank is

(A) $\sqrt{\frac{Km}{(M+m)}}\,\ell$ (B) $\sqrt{\frac{m}{(M+m)}}\,\ell$

(C) $\sqrt{\frac{Km}{M(M+m)}}\,\ell$ (D) $\sqrt{\frac{Km}{(M+m)}}\,\ell$

18. Maximum kinetic energy of block m is

(A) $\frac{Km^2}{2M(M+m)}$ (B) $\frac{Km^2}{M(M+m)}$

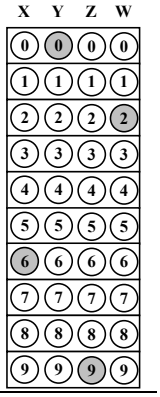
(C) $\frac{KM\ell^2}{2(M+m)}$ (D) none of these

Rough work

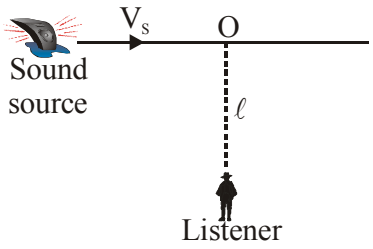
SECTION – C

Integer Answer Type

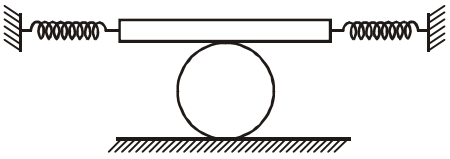
This section contains 10 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



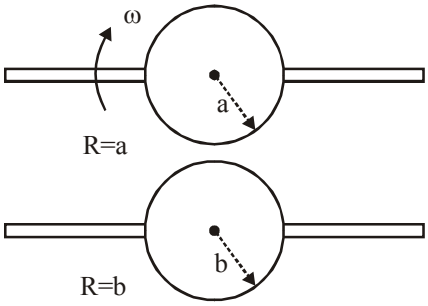
1. A source of sound with natural frequency F_0 moves uniformly along a line separated from a stationary listener by a distance $\ell = 42\sqrt{5}$ m. The velocity of source is equal to n = 0.5 fraction of velocity of sound. If the distance between source and listener at the moment when the listener receives actual frequency of the source is (21 × n) meter, find the value of n.



2. In the figure shown, mass of the plank is m and that of the solid cylinder is 8m. Springs are light. The plank is slightly displaced from equilibrium and then released. Find the period of small oscillations (in seconds) of the plank. There is no slipping at any contact point. The ratio of the mass of the plank and stiffness of the spring i.e. $\frac{m}{K} = \frac{2}{\pi^2}$.



3. Two uniform spheres shown in figure rotate separately on parallel axis. The upper sphere has angular speed ω_0 and lower sphere is at rest. Now the two spheres are moved together so that their surfaces touch. After a short time the two spheres now in contact are rotating without slipping. Find the final rate of rotation of upper sphere in rad/s. [given moment of inertia of upper sphere I_1 = 4 kg m^2, moment of inertia of lower sphere I_2 = 2 kg m^2, a = 2 m, b = 1 m, ω_0 = 12 rad/s].



Rough work

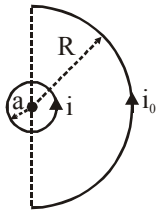
4 For an equilateral prism, it is observed that when a ray strikes grazingly at one face it emerges grazingly at the other. Find the refractive index of the prism

5. Consider a hydrogen like atom whose energy in n^{th} excited state is given by

$$E_n = -\frac{13.6}{n^2}Z^2$$

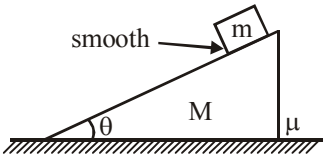
when this excited atom makes a transition from an excited state to ground state. The most energetic photons have energy E_{max} = 52.224 eV and the least energetic photons have energy E_{min} = 1.224 eV. Find the atomic number of atom.

6. A small current carrying loop having current i is placed in the plane of paper as shown. Another semicircular loop having current i_0 is placed concentrically in the same plane as that of small loop, the radius of semicircular loop is R (R >> a). Find the force applied by the smaller ring on bigger ring in newton.



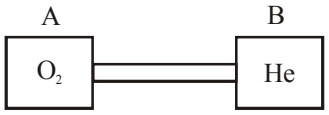
(Given R = 1 m, $i = i_0 = \frac{40}{\sqrt{\mu_0}}$ A , a = 0.1 m)

7. In figure if μ = 0.1 is the coefficient of friction between the wedge and the horizontal surface then for what maximum value of m in kg, the wedge remains at rest [take M = 18 kg and θ = 45º].



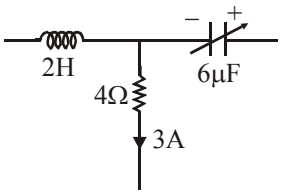
Rough work

8. Two containers A and B are connected by a conducting solid cylindrical rod of length $\frac{242}{7}$ cm and radius $\sqrt{8.3}$ cm. Thermal conductivity of the rod is 693 watt/mole–K. The container A contains two mole of oxygen gas and the container B contains four mole of helium gas. At time t = 0 temperature difference of the containers is 50ºC, after what time (in seconds) temperature difference between them will be 25ºC. Transfer of heat takes place through the rod only. Neglect radiation loss. Take R = 8.3 J/mole–K and $\pi = \frac{22}{7}$.



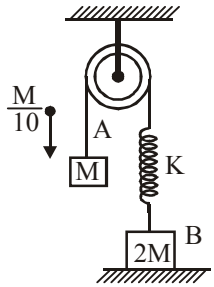
9. The figure shows part of a bigger circuit. The capacity of the capacitor is 6μF and decreasing at the constant rate 0.5 μF s^{-1}. The potential difference across the capacitor is changing as under

$\frac{dV}{dt} = 2Vs^{-1}$, $\frac{d^2V}{dt^2} = \frac{1}{2}V s^{-2}$.



The current in the 4Ω resistor is decreasing at the rate of 1 μA s^{-1}. What is the potential difference (in micro volts) across the inductor at this moment ?

10. In the shown figure, a particle of mass $\frac{M}{10}$ strikes the block of mass M with velocity v_0 and gets attached to it. For what velocity v_0 (in ms^{-1}), the block B just able to leave the ground ?
(Given M = 100 gm, K = 880 N/m)



Rough work

Chemistry PART – II

SECTION – A

Straight Objective Type

This section contains 8 multiple choice questions numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. $\underset{(1\,lit)}{Ag\,|\,Ag^{+}\,(1\,M)}\,||\,\underset{(1\,lit)}{Ag^{+}\,(2\,M)}\,|\,Ag$

0.5 F of electricity in anode and 1 F in the cathode is first passed making them independent electrolytic cell at 298 K. EMF of the cell after electrolysis will be:

(A) increased (B) decreased
(C) no change (D) time is also required to find it

2. For two gases A and B with molecular weights M_A and M_B. It is observed that at a certain temperature T, the average speed of A is equal to root mean square speed of B. Thus the average speed of A can be made equal to the average speed of B

State I : By keeping A at higher temperature than B.
State II : By keeping B at higher temperature.
State III : By keeping A at $\frac{8}{3\pi}$ times temperature of B.

Correct statement is/are:

(A) both State I and State II (B) both State II and State III
(C) both State I and State III (D) only State I

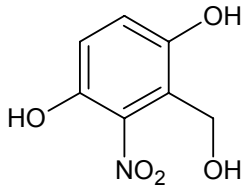
3. Consider the equilibrium:

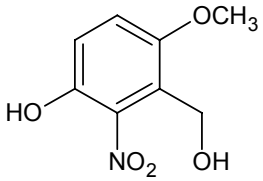
$2CO(g) + O_2 \rightleftharpoons 2CO_2(g) + \text{heat}$

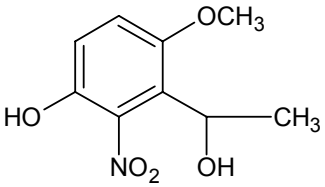
If O_2 is added and volume of the reaction vessel is increased, the equilibrium will

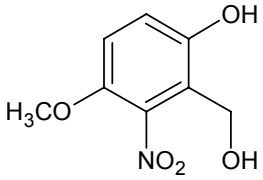
(A) shift in forward direction (B) shift in backward direction
(C) remain unchanged (D) be unpredictable

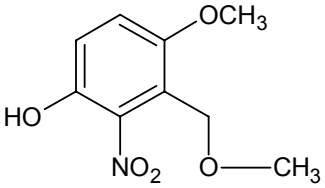
Rough Work

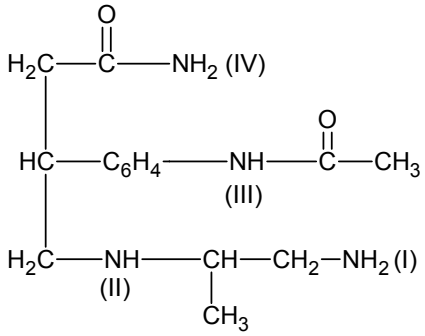
4.  $\xrightarrow[(1\,eq.)]{CH_2N_2}$ A, A is :

(A) 

(B) 

(C) 

(D) 

5. 

Arrange the above mentioned nitrogen atoms of the compound in decreasing order of basic nature?

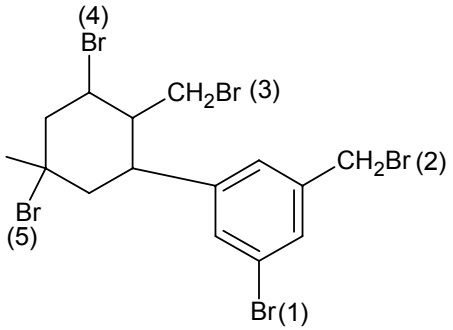
(A) I > II > IV > III
(B) II > I > IV > III
(C) II > I > III > IV
(D) I > IV > III > II

6. Which of the following compounds can form silicones on hydrolysis?

(A) $(CH_3)_2SiCl_2$
(B) $(SiH_3)_3N$
(C) $SiCl_4$
(D) All

Rough Work

7.



Arrange the above mentioned bromide atoms in decreasing order of their reactivity towards hydrolysis through S_N1 path

(A) 2 > 1 > 5 > 4 > 3 (B) 5 > 2 > 1 > 3 > 4

(C) 2 > 5 > 4 > 3 > 1 (D) 5 > 4 > 2 > 1 > 3

8. Gold number gives

(A) amount of gold in colloid

(B) weight in mg of colloid required to break gold sol.

(C) weight in mg of colloid to protect gold sol.

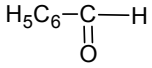
(D) amount of NaCl required to coagulate gold sol.

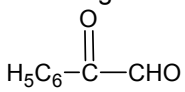
Rough Work

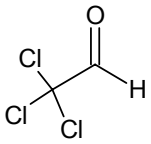
Multiple Correct Choice Type

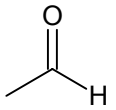
This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. Which of the following compounds will disproportionate on treating with conc. NaOH?

(A) 

(B) 

(C) 

(D) 

10. If electrolysis of 1 M $NiSO_4$ is carried with inert electrodes at pH = 7 at 25°C (the pressure of gases are taken 1 bar).

$E^{o}_{Ni^{2+}/Ni} = -0.25$ V $E^{o}_{H_2O/H_2} = -0.83$ V

The product of electrolysis at cathode and at anode (initially) are:

(A) Ni at cathode
(B) H_2 at cathode
(C) O_2 at anodes
(D) $S_2O_8^{2-}$ at anode

11. Which of the following is/are correct

(A) $[CrCl_2(CN)_2(NH_3)_2]$ and $[CrCl_3(NH_3)_3]$, both have d^2sp^3 hybridisation
(B) Magnetic moment of $[PdCl_4]^{2-}$ is zero
(C) $[Cu(NH_3)_4]^{2+}$, $[Pt(NH_3)_4]^{2+}$ and $[Ni(CN)_4]^{2-}$, all have dsp^2 hybridization of central metal
(D) It is difficult to substitute chelating ligands compared to similar monodentate ligands

12. Which of the following is/are correct?

(A) On strong heating $FeCl_3$ gives $FeCl_2$ and Cl_2
(B) Dilute $CuCl_2$ is blue due to formation of $[CuCl_4]^{2-}$
(C) PH_3 can be obtained by heating red P with NaOH
(D) PH_3, H_3PO_3 and H_3PO_2, all are reducing agents

13. $2X(g) + Y(g) + 3Z(g) \longrightarrow$ Products

The rate equation of the above reaction is given by:

Rate = $K[X][Y]^{o}[Z]^{2}$

Choose correct statements:

(A) the rate of reaction becomes doubled by doubling the conc. of (X)
(B) rate of reaction decreases by reducing the conc. of (Y) to half of the original value
(C) the half life of (Z) increases by increasing its conc.
(D) if 75% of (X) undergoes reaction in 20 sec, 50% of (X) will react in 10 sec

Rough Work

PARAGRAPH TYPE

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph for Question Nos. 14 to 15

There are 4 glass bottles containing water solutions. These bottles are not labeled. Bottles 1,2,3 Contained colourless solutions, while bottle 4 contained a blue solution. Each bottle is having any one solution among the following (Each bottle is having different solution).
Copper (III) sulphate
Sodium carbonates
Lead nitrate
Hydrochloric acid
By mixing samples of the contents of the bottles, in pairs, following observations are made:

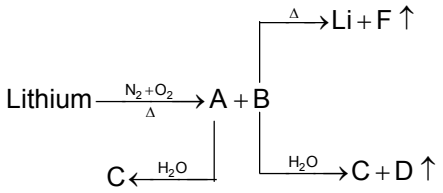
(1)	Bottle 1 + Bottle 2	White ppt.
(2)	Bottle 1 + Bottle 3	White ppt.
(3)	Bottle 1 + Bottle 2	White ppt.
(4)	Bottle 2 + Bottle 3	Colourless gas evolved
(5)	Bottle 2 + Bottle 4	No visible reaction
(6)	Bottle 3 + Bottle 4	Blue ppt.

14. Colourless solution present in Bottle 1 is :
(A) $CuSO_4$ (B) $Pb(NO_3)_2$
(C) HCl (D) Na_2CO_3

15. Nature of gas evolved in observation (4) is:
(A) Acidic (B) Neutral
(C) Basic (D) Amphoteric

Rough Work

Paragraph for Question Nos. 16 to 18



Answer, the following questions:

16. Which of the following is compound (A)?
(A) Li_3N (B) Li_2O
(C) Li_2O_2 (D) LiO_2

17. The anion present in compound (C) is:
(A) O^{2-} (B) N^{3-}
(C) OH^- (D) NO_3^-

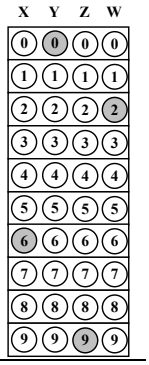
18. Which of the following is a property of gas F↑?
(A) It is paramagnetic in nature
(B) It's bond order is 3
(C) It does not contain lone pairs
(D) It contains two sigma and one pi bond per molecule

Rough Work

SECTION – C

Integer Answer Type

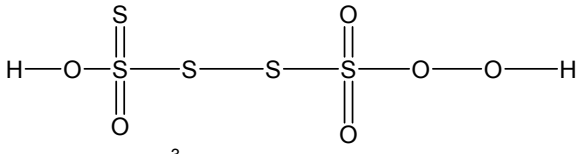
This section contains 10 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. A compound (A) having molecular formula $C_7H_{11}Br$ is optically active. A reacts with HBr in the absence of peroxide to give isomeric products (B) and (C). Treating (A) with potassium-t-butoxide gives (D). (D) on reductive ozonolysis gives two moles of formaldehyde and one mole of 1, 3 cyclopentane dione. (A) in the presence of peroxide reacts with HBr to give (E). The total number of chiral carbon in one molecule of (E) is:

2. How many maximum number of e^- of an atom will have the following set of quantum numbers

   $n = 4, \ell = 0,1,2 \quad m = +2,-2, \quad S = \pm\frac{1}{2}$

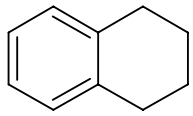
3. 

   How many sp^3 hybridized atoms are present in one molecule of the above hypothetical compound?

4. A complex of potassium, iron and cyanide ions $K_x[Fe(CN)_y]$ is 100% ionized at 1 molal. If its elevation in boiling points is 2.08^oC, find out the value of 'x' in the complex: (Given $K_b = 0.52^oC\ mol^{-1}$)

5. Phosphorus sulphide P_4S_3, a well known chemical used in match industry has how may P – S bonds?

Rough Work

6. 
$\xrightarrow[\text{(excess)}]{\text{NBS}} (x) \xrightarrow[\text{(excess)}]{\text{KOH/H}_2\text{O}} (y)_{\text{unstable}} \xrightarrow{\Delta} z.$

The sp^2-hybridised carbon atoms present in one molecule of the end product z is:

7. Find $10 \times \log_{10} K$ for the reaction at 298 K.

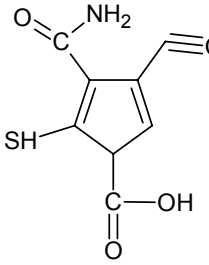
$2H^+(aq) + D_2(g) \rightleftharpoons H_2(g) + 2D^+(aq)$

Given $E^{\circ}_{D^+/D_2} = -2.955$ mv

Take $\left(\frac{RT2.303}{F}\right) = 0.0591$

8. Wavelength of the second line of Balmer series of a hydrogen like species is $\frac{4}{3R}$, what is the atomic number of the hydrogen like species? [R = Rydberg's constant)

9. What is the number of moles of CH_4 formed in the given reaction,



$\xrightarrow[\text{excess}]{CH_3MgBr}$ Number of moles of CH_4

10. No. of geometrical isomers for $[M(gly)Cl_2Br_2]^+$ is:

Rough Work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 8 multiple choice questions numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If k be a positive integer, then the number of values of k satisfying $\int_0^{\pi/2}\left[k^2\left(\cos 3x+\frac{1}{2}\cos x\right)+\sin x-2k\cos x\right]dx\le 1$, is

(A) 10 (B) 11
(C) 12 (D) none of these

2. If $I_1=\int_{-100}^{101}\frac{dx}{(5+2x-2x^2)(1+e^{2-4x})}$ and $I_2=\int_{-100}^{101}\frac{dx}{5+2x-2x^2}$, then $\frac{I_1}{I_2}$ is

(A) $\frac{1}{2}$ (B) 2
(C) 0 (D) none of these

3. If $x\,f(x)=3(f(x))^2+2$ then $\int\frac{2x^2-12xf(x)+f(x)}{(6f(x)-x)(x^2-(f(x))^2}\,dx$ is equal to

(A) $\frac{1}{(6f(x)-x)^2}+c$ (B) $\frac{1}{x^2-f(x)}+c$
(C) $\frac{1}{x^2+xf(x)}+c$ (D) none of these

4. Let $f(x)=\lim_{h\to 0}\left(\frac{(\sin(x+h))^{\ln(x+h)}-(\sin x)^{\ln x}}{h}\right)$ then $f\left(\frac{\pi}{2}\right)$ is

(A) 0 (B) 1
(C) $\ln\left(\frac{\pi}{2}\right)$ (D) none of these

Rough work

5. Number of 4 digit number of the form $N = x_1\ x_2\ x_3\ x_4$ such that $2000 \le N \le 5000$, $4 \le x_2 < x_3 \le 8$ and N is multiple of 5 is/are
(A) 80 (B) 60
(C) 40 (D) 30

6. A biased coin is tossed repeatedly until a tail appears for the 1st time. The head is 2 times likely to appear as tail. The probability that the number of tosses required will be more than 6 given that in 1st three toss no tail has occured is/are
(A) $\frac{16}{81}$ (B) $\frac{32}{243}$
(C) $\frac{64}{729}$ (D) none of these

7. The product of all the even divisor of N = 9000 is
(A) 721×10^5 (B) 726×10^6
(C) 729×10^6 (D) none of these

8. Let the matrices $A = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}$ and P be any orthogonal matrix such that Q = PAP′ and let $R = \left[r_{ij}\right]_{2\times 2} = P'Q^8P$ then
(A) $r_{11} = 81$ (B) $r_{11} = 81\sqrt{3}$
(C) $r_{11} = 4\sqrt{3}$ (D) $r_{11} = -\sqrt{3}$

Multiple Correct Choice Type

This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. The radii of the escribed circle of ΔABC are r_1, r_2 and r_3 respectively (opposite to vertices A, B and C). If $r_1 + r_2 = 3R$ and $r_2 + r_3 = 2R$ then
(A) $\angle A = 90^\circ$ (B) $\angle B = 45^\circ$
(C) $\angle C = 60^\circ$ (D) right angle isoscelles triangle

Rough work

10. Let AB : y = x be a chord to the parabola $y^2 + 4x = 4$ and P be any variable point on the ellipse $x^2 + 9(y - 4)^2 - 9 = 0$. If Δ_1 and Δ_2 be the minimum and maximum area of triangle ABP respectively then

(A) $\Delta_2 = 2\sqrt{2}\left(4 + \sqrt{10}\right)$ (B) $\Delta_2 - \Delta_1 = 8\sqrt{5}$

(C) $\frac{\Delta_2}{\Delta_1} = \frac{13 + 4\sqrt{10}}{3}$ (D) none of these

11. The all possible order pairs (a, b) that satisfy the relation

$(4 - 3i)a^2 + (3 + 2i)ab = 4b^2 - \frac{a^2}{2} + (3ab - 2b^2)i$ is/are

(A) (4, –3) (B) (6, 9)
(C) (10, 15) (D) none of these

12. Let $f(x) = \begin{cases} x + 1 & , \ 0 \le x \le 1 \\ 2x^2 - 6x + 6 & , \ 1 \le x \le 2 \end{cases}$

And $g(t) = \int_{t-1}^{t} f(x)dx$, $t \in [1, 2]$. Which of the following is true statement

(A) f(x) is continuous and differentiable $\forall$ $x \in [0, 2]$

(B) g(t) is maximum at $t = \frac{3}{2}$

(C) g(t) is minimum at t = 2

(D) all the above

13. A sq. matrix A of order 2 consisting element 0, 1, 2 and 4 then possible value of det(A) is/are

(A) 2 (B) 4
(C) 8 (D) none of these

Rough work

PARAGRAPH TYPE

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph for Question Nos. 14 to 15

Read the following write up carefully and answer the following questions:

Let B_n denotes the event that n fair dice are rolled once with $P(B_n) = \frac{1}{2^n}$, $n \in N$. e.g. $P(B_1) = \frac{1}{2}$, $P(B_2) = \frac{1}{2^2}$, $P(B_3) = \frac{1}{2^3}$, ….. and $P(B_n) = \frac{1}{2^n}$. Hence B_1, B_2, B_3, ….. B_n are pairwise mutually exclusive and exhaustive events as $n \to \infty$. The event A occurs with atleast one of the event B_1, B_2, ….., B_n and denotes that the sum of the numbers appearing on the dice is S.

14. If even number of dice has been rolled, the probability that S = 4, is

(A) very closed to $\frac{1}{2}$ (B) very closed to $\frac{1}{4}$

(C) very closed to $\frac{1}{8}$ (D) very closed to $\frac{1}{16}$

15. Probability that greatest number on the dice is 4 if three dice are known to have been rolled, is

(A) $\frac{37}{216}$ (B) $\frac{64}{216}$

(C) $\frac{27}{216}$ (D) $\frac{31}{216}$

Rough work

Paragraph for Question Nos. 16 to 18

Read the following write up carefully and answer the following questions:
Tangents are drawn to the parabola $y^2 = 4x$ from the point P(6, 5) to touch the parabola at Q and R. C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R. Both the circles C_1 and C_2 pass through the focus of the parabola.

16. Area of the ΔPQR equals
(A) 1/2 (B) 1
(C) 2 (D) 1/4

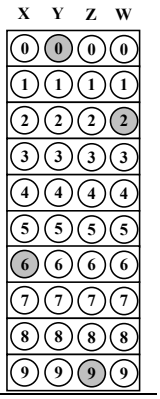
17. Radius of the circle C_2 is
(A) $5\sqrt{5}$ (B) $5\sqrt{10}$
(C) $10\sqrt{2}$ (D) $\sqrt{210}$

18. The common chord of the circles C_1 and C_2 passes through the
(A) incentre (B) circumcentre
(C) centroid (D) orthocentre of the ΔPQR

SECTION – C

Integer Answer Type

This section contains 10 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. If 3n be the sum of all possible integer values of 'p' for which $\left|x+\frac{1}{x}-3\right| = p-3$ has exactly two distinct solutions then the value of n is _____.

Rough work

2. For positive t, let α, β be the root of the quadratic equation $x^2 + t^2x - 2t = 0$. If minimum value of $\int_{-1}^{2}\left(\left(x+\frac{1}{\alpha^2}\right)\left(x+\frac{1}{\beta^2}+\frac{1}{\alpha\beta}\right)\right)dx$ is $\sqrt{p}+q$ where p is a rational number and $q \in N$, then the value of [p] + q is (where [.] denotes G.I.F)

3. A species has an initial population 4^{10}. At the end of the 1st day, the population increases by 50%. At the end of the 2nd day it decreases by the same percentage. If the process continues in the same pattern, and let 5x be the number of days required for population to be 3^{10}, then x is/are _____

4. The equation $2x^3 - 3x^2 + p = 0$ has 3 distinct real roots then [p] is _____. (Where [.] denotes the greatest integer function).

5. A plane P passes through the intersection of $\vec{r}\cdot\left(\hat{i}+\hat{j}+\hat{k}\right)+3=0$ and $\vec{r}\cdot\left(\hat{i}-\hat{j}+\hat{k}\right)=2$. If p divide the line segment joining (3, 0, 2) and (0, 3, –1) in the ratio 2 : 1 internally and the equation of plane is given by $\vec{r}\cdot\left(a\hat{i}-\hat{j}+b\hat{k}\right)=c$ when $a, b, c \in N$ than a + b + c is/are _____.

6. If the distance from (k, 0), k > 0 to the tangent to the parabola $y^2 = 4x$ at $\left(\frac{1}{4}, 1\right)$ is $2\sqrt{5}$ then the value of [k] is _____. (Where [.] denotes the greatest integer function)

7. Consider a polynomial P(x) of the least degree that has a maximum equal to 6 at x = 1, and a minimum equal to 2 at x = 3. Let P(2) + P'(0) is equal to n then $\left[\frac{n}{2}\right]$ is _____.

8. Let y = g(x) be the image of f(x) = x + sin x about the line x + y = 0. If the area bounded by y = g(x),
x-axis, x = 0 and $x = 2\pi$ is A, then $\frac{A}{\pi^2}$ is _____.

9. If $I_n = \int_0^{\infty} e^{-x}(\sin x)^n dx$ (n > 1), then the value of $\frac{101\, I_{10}}{10\, I_8}$ is equal to _____.

10. If $|\vec{a}| = 2$, $|\vec{b}| = 3$, and $|\vec{c}| = 4$ and maximum of $|\vec{a}-\vec{b}|^2 + |\vec{b}-\vec{c}|^2 + |\vec{c}-\vec{a}|^2$ = 10 p + q where p, q $\in$ {0, 1, 2, 3, ….. 9} then |p – q| is _____

Rough work